

Donghyung Lee

DevOps Engineer

E-mail: genius5711@gmail.com Blog(KR): somaz.tistory.com Blog(EN): somaz.blog GitHub: github.com/somaz94

Introduction

Hello, I am Donghyung Lee, a DevOps Engineer.

With a goal of maintaining 99.9% infrastructure availability, I have maximized operational efficiency by establishing processes for codifying all changes (IaC) and establishing systematic incident documentation (postmortem) processes. My greatest strength is the habit of thinking beyond simple operations, contemplating the meaning and purpose of each task, and seeking better directions.

I have experience in designing and building cloud infrastructure on AWS and GCP, and am proficient in service operations in Kubernetes (EKS, GKE, on-premises) environments, IaC implementation using Terraform and Ansible, and CI/CD pipeline construction with GitLab CI/CD, GitHub Actions, ArgoCD, and Jenkins. I also have experience building monitoring and logging systems based on PLG Stack and ELK Stack, and have strengths in operations automation using Python and Bash.

Across 3 companies, I have independently or proactively led cloud migration (AWS → GCP), hybrid architecture design, and on-premises PaaS solution construction, achieving quantitative results including 20% AWS cost reduction through hybrid architecture transition and 50% reduction through GCP migration resource optimization, 50-67% deployment time reduction, and 90%+ manual work automation.

I aim to continue contributing to infrastructure architecture advancement and organizational productivity improvement.

Strengths

Quick Understanding & Adaptation

- Quickly adapt to new environments and tools, contributing immediately to diverse projects

Systematic Documentation & Knowledge Sharing

- Document all work processes and share with team, contributing to organizational knowledge assets

Record-Based Continuous Improvement

- Manage all tasks including incident response, infra changes, and troubleshooting through records to prevent recurrence and drive improvement

Full-Stack Infrastructure Experience

- End-to-end infrastructure experience spanning cloud (AWS, GCP), on-premises, networking, CI/CD, and monitoring

Skills

Cloud	AWS, GCP
Container Orchestration	Kubernetes (EKS/GKE Multi-cluster Ops, On-premises HA with CNI/CSI Optimization)
Infrastructure as Code	Terraform, Ansible, Vagrant, Packer
CI/CD	GitHub Actions, GitLab CI/CD, ArgoCD, Jenkins
Monitoring & Logging	PLG Stack, ELK Stack, EFK Stack, ECK Operator, Thanos, Vaultwarden
Automation	Go, Python, Bash Scripting
Database	MySQL, PostgreSQL, Redis, MongoDB
OS & Servers	Ubuntu, Rocky Linux, CentOS, Debian, GPU Server

Professional Experience

Phantom (Concrit Studio)

2024.10 ~ Present

DevOps Engineer

Dedicated DevOps infrastructure engineer designing, building, and operating AWS-based cloud and on-premises servers. Achieved both cost optimization and operational stability through hybrid cloud architecture design.

- AWS-on-premises hybrid cloud architecture design and construction
- GitOps CI/CD pipeline construction and enhancement based on GitLab CI / ArgoCD / Jenkins
- Built ELK Stack logging → migrated to EFK (70% resource reduction) → moved to ECK Operator + Elastic Stack 9.0 CRD-based declarative management (automatic TLS rotation, one-line CR spec.version rolling upgrades)
- Monitoring enhancement with kube-prometheus-stack (18 alert rules, 10 custom dashboards, 4 DB exporters)
- K8s cluster upgrade automation via Kubespray (9-step health checks, etcd backup/restore)
- Vaultwarden password management system (GitLab SSO, automated backup, team access control)
- GitLab-Google Drive document automation system
- Internal LLM service deployment (Ollama + Open WebUI)
- Slack APK distribution automation bot development and operation
- GitLab, NAS, SVN account management automation via Bash Scripts
- Integrated application login environment using GitLab SSO
- Kubernetes storage system (NFS dynamic provisioning + Local Path Provisioner + in-house open-source static file server static-file-server)

Key Achievements

- 20% Monthly Infra Cost Reduction (\$60,000 → \$48,000, ~\$144K annual savings → reinvested in new projects)
 - 67% Deployment Time Reduction (30min → 10min, 3x weekly deployments for faster market response)
 - 90% Deployment Resource Efficiency (Enabling dev team to focus on core development)
 - 90% Log Search Time Reduction
 - 70% MTTR Improvement (99.9% Service Availability)
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NerdyStar

2023.03 ~ 2024.07

DevOps Engineer

Led the large-scale AWS → GCP cloud migration at a game and blockchain startup, and rebuilt GitOps-based CI/CD pipelines.

- AWS → GCP large-scale cloud migration infrastructure design and construction
- GitOps CI/CD pipeline construction based on GitLab CI / ArgoCD
- Certificate automation using Cert-Manager and Let's Encrypt
- Game/blockchain data collection automation via Cloud Functions + Cloud Scheduler (Google Sheets API integration)
- Integrated monitoring system based on PLG Stack (Prometheus, Loki, Grafana)
- GPU server infrastructure design and construction for AI/ML workloads

Key Achievements

- 50% Cloud Cost Reduction (\$10,000 → \$5,000/month)
 - 50% Deployment Time Reduction (40min → 20min)
 - 95% Rollback Time Reduction (30min → 1-2min)
 - 95% Data Collection Automation (0% data gap rate)
 - 99.9% Service Availability (99.5% → 99.9%)
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Iorchard

2022.04 ~ 2023.03

Infra Engineer

Built PaaS (Kubernetes + OpenStack + Ceph) infrastructure solutions for clients and provided technical support. Designed and built on-premises cloud infrastructure for financial and public sector clients, and conducted operations training.

- Customized PaaS solution design and construction (Kubernetes HA + OpenStack + Ceph)
- DP (Delivery Partner) Kubernetes operations training program design and delivery (6 months)
- Kubernetes certificate validity extension automation script (1yr → 10yr)
- OpenStack and Ceph storage cluster operations and optimization
- Regular cluster inspections and troubleshooting
- Ansible-based server provisioning automation eliminating configuration drift issues

Key Achievements

- 5 Cases Client PaaS Solution Deployments (99.9% Availability)
 - 10x Certificate Renewal Cycle Extension (Annual → Once per 10 years)
 - 50hrs Annual Ops Savings (Across 10 clients)
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ETech System

2022.01 ~ 2022.04

System Engineer

- Hardware Server and SAN storage construction
- Transitioned to DevOps field for career growth

Opensource Projects

Kubernetes CRD

[k8s-namespace-sync](#) — Automatically sync Kubernetes namespace resources across clusters

[helios-lb](#) — Custom load balancer controller for Kubernetes

[network-policy-generator](#) — Kubernetes NetworkPolicy auto-generation controller

GitHub Actions

[Compress Decompress Action](#) — Compress/decompress files in GitHub Actions workflows

[Image Tag Updater](#) — Automatically update image tags in YAML or Helm files

[Go Git Commit Action](#) — Go-based Git commit creation and push

[Extract Commit Action](#) — Extract Git commit SHA, message, and author information

[Multi Git Mirror Action](#) — Automate mirroring across multiple Git repositories

[Kube Diff Action](#) — Compare and report Kubernetes resource changes

Ansible Galaxy

[Ansible K8s IAC Tool](#) — Automated installation collection for Kubernetes and IaC tools

[Ansible Kubectl Krew](#) — kubectl plugin Krew management role

[Ansible User Management](#) — Linux user and SSH key management role

DevOps Tools

[bash-pilot](#) — AI-powered Bash command assistant CLI tool

[git-bridge](#) — Git repository mirroring and sync CLI tool

[static-file-server](#) — Go-based static file server with directory UI, file preview, search, ZIP batch download

Chrome Extensions

[Dev Toolkit](#) — A collection of useful developer utilities

[QRify](#) — Quickly generate QR codes for URLs or text

Education

Chungnam National University - Public Administration (Transfer)

2018.03 ~ 2020.08

National Institute for Lifelong Education - Business Administration (Academic Credit Bank)

2016.09 ~ 2018.02

Certifications

Engineer Information Processing

2021.11

AWS Certified Solution Architect - Associate (Expired)

2021.11

Network Administrator Level 2

2021.10

Linux Master Level 2

2021.10

Computer Specialist Level 1

2021.04

TOEIC Speaking Test - 130/Intermediate Mid 3 (Expired)

2021.02

Study

[T101\(Terraform\) Study](#)

2023.08 ~ 2023.10

[AEWS\(EKS\) Study](#)

2023.04 ~ 2023.06

[PKOS\(Kubernetes\) Study](#)

2023.01 ~ 2023.02

[Cloud Security Training](#)

2022.02 ~ 2022.04

Security Solution Operation Specialist Training Program

2021.07 ~ 2021.12